

Investigating Algorithmic Bias' Effect on Social Media Content Curation

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Abstract

Algorithmic bias is a prevalent topic in today's society that users often encounter several times on a daily basis, especially given the rise of social media and digital content. "Shadowbanning" is a concept that will be explored, with the idea that certain topics are more likely to be shown than others. Given this, the paper investigates how algorithmic bias can potentially hide content from users, based on the topic.

The data being explored are Tweets from USC students during the 2024 election. The tweets are embedded and clustered. The clusters are then mapped to topics and segmented by weeks. From each week, the key metric is extracted: views/likes where a lower ratio could indicate the presence of shadow-banning. The data can then be analyzed on a week-by-week basis, as well as on a topic-by-topic basis. For each topic, the actual distribution of the key metric can be compared against a theoretical Gaussian distribution to determine the presence of bias. Other statistical features such as completing a t-test or calculating p-values can be found to support our analysis.

[insert results and conclusion in the future]

1 Introduction

Social media platforms are a central part of many users' lives, delivering large volumes of information and content at high speed. A way to manage and direct what content reaches which audience, is by combining content moderation and artificial intelligence, a practice known as AI-powered content moderation. This research paper investigates algorithmic bias found in AI-powered content moderation and what factors may influence this.

Literature on this topic suggests that AI has been used to suppress or amplify certain demographics on similar platforms such as Youtube, Facebook, and Twitter. For example, algorithmic content moderation results in more left-center aligned political articles being shared on the Bluesky, a social media platform similar to Twitter (Quelle, 2024). Similarly, research has shown content moderation can be curated to create echo chambers Wang et al. (2024), which

could perpetuate confirmation bias amongst its users by delivering content with similar narratives or themes (Huszar et al., 2020).

1.1 Research Question

To what extent does algorithmic bias influence the content curation process on social media platforms? And can we quantify differences in the ways content of varying ideologies are spread and potentially suppressed across these platforms?

1.2 Hypothesis

Intersectional factors such as race, political affiliation, and gender influence the algorithmic bias present in the content curation process, specifically censoring marginalized voices and certain political ideologies.

2 Related Works

There has been an abundance of adjacent research done on the topic of algorithmic bias on content curation as it is a popular topic given the current technological and political landscape. The first set of papers explores the effects of algorithmic bias on content moderation, while the second set explores the shortcomings of algorithmic content moderation and its adverse effects. Binns et al. (2017) explores ways to measure the inherent bias that algorithmic content moderation systems possess, specifically content labeled by either males or females. Quelle (2024) discusses the decentralized social media platform: Bluesky and how its network structure, user iterations, and political polarization influences the content curation process. Wang et al. (2024) investigates the presence of "echo chambers" and how algorithmic design can recommend content that of similar themes, possibly strengthening previous beliefs in the user. Similarly, Huszar et al. (2020) discusses how algorithmic bias can amplify or silence certain political views.

The following research explore the difficulties of scaling content moderation, as social platforms such as Youtube, Facebook, and X are scaling at extreme speeds. Gorwa et al. (2020) looks at how these large social platforms' reliance on algorithmic content moderation, although effective, can lead to problems relating to transparency, fairness, and political messaging. The difficulties of content moderation due to algorithmic bias are further explored by Gongane et al. (2022), explaining the difficulties of moderating content such as fake news, hate speech, and cyberbullying. Lastly, Young (2022) explores how content moderation may be too limiting too, which exacerbates the difficulties of catering content too strongly to certain audiences and leading to potential real-life repercussions.

3 Methodology

3.1 Overview

We will primarily explore the idea of "shadowbanning," where a tweet may have relatively high levels of engagement through replies, retweets, and likes, but comparatively low views. It's a simple metric but its inferential value also derives from the exact ways we linearly combine these factors to derive a numeric measure, the movement of which we can also study as its own phenomena. We can then apply this across a segmentation of topics and accounts, analyzing the demographics of the accounts posting to identify gaps in these engagement metrics relative to these demographic factors. We can also track and regress against the general demographic engagement metrics of different identified topics to see there is any correlative motion between the algorithmic suppression of any particular topic based on the demographics of the conversant accounts, engaging in analysis of both the populations whom are the topic of and engage in particular domains of discourse.

We plan to utilize the USC X 2024 Election dataset Balasubramanian et al. (2024), which contains millions of tweets related to the 2024 election. We will need to perform agglomerative clustering alongside keyword identification initially to identify a list of areas of import, and we can then perform our engagement analyses. We seek to use rules-based natural language processing wherever possible, but for more subjective resolutions like political and racial classifications, we will have to rely on and hold to account the biases of black-box methods like embedding models and transformers.

3.2 Technical Implementation

The technical implementation begins by embedding the USC X 2024 Election dataset for processing and clustering the embeddings. Then these clusters are mapped to the topics. These topic-separated clusters are then split up by week and formatted as a .csv file for easy processing.

The next portion of analysis focuses on analyzing the key metric (views/likes) in two methods: as a timeseries and in a distribution. The first facet of analysis will track the key metric per topic across each week. The second facet of analysis investigates for each topic, how the view/like ratio for each tweet compares against a theoretical Gaussian distribution. This theoretical Gaussian distribution will be created by taking the topic's mean and standard deviation of the key metric. A t-test will be conducted on each point versus the mean, to determine whether there's a great enough difference. This will determine the presence of shadow-banning for each topic.

4 Results

The results from a single week of analysis is shown below.

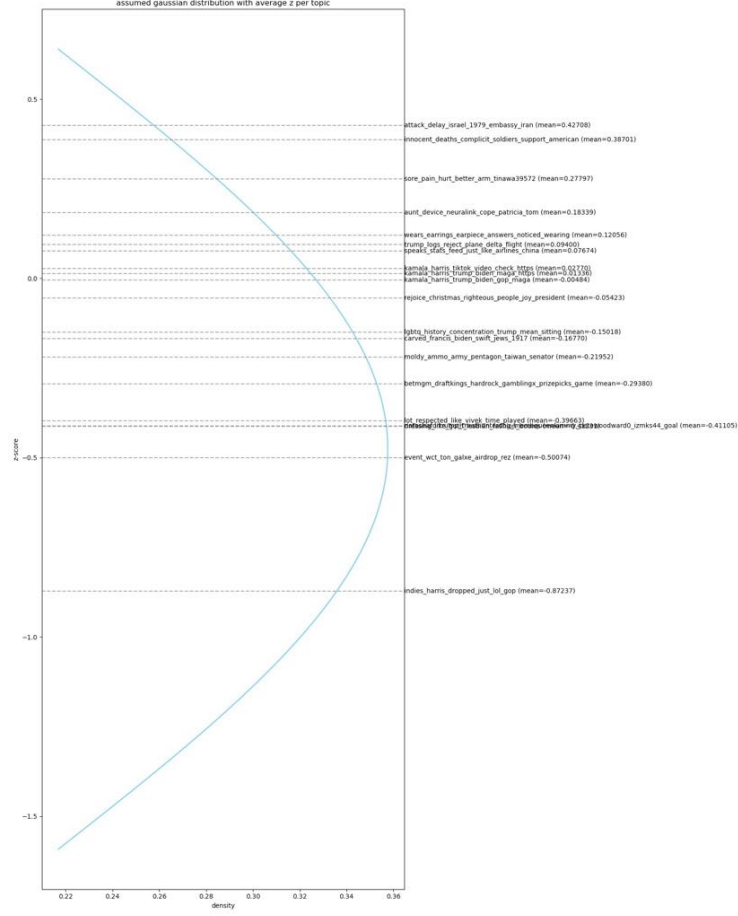


Figure 1: View/Like Ratio for each clustered topic, plotted against a theoretical Gaussian distribution - for a single week

5 Discussion

This section will be populated once the analysis has been completed. In the meantime, the results for the midterm report are included above.

References

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